Declared truncation v0.1 - reconstruction convention

Status: **new, declared reconstruction basis**. It is not a transcription of a historical 41-coupling calculation.

Fields and conventions

Euclidean background-field FRG in four dimensions with fields `g_mu_nu`, Standard Model fields, and real gauge-singlet scalars `Phi_c` and `E`. The flow is the Wetterich equation

```
partial_t Gamma_k = 1/2 STr[(Gamma_k^(2) + R_k)^(-1) partial_t R_k].
```

Use background-covariant gauge fixing and Faddeev-Popov ghosts. A regulator choice, gauge parameters, and projection backgrounds must be recorded with every numerical result; no coefficient in this project is valid without them.

Coordinate basis

The 41 coordinates follow `model.NAMES` in order:

1. `gN`: dimensionless Newton coupling; Einstein-Hilbert `R`.
2. `lambdaN`: dimensionless cosmological term.
3. `a_R2`: `R^2`.
4. `a_C2`: Weyl-tensor squared. 5-7. `g1`, `g2`, `g3`: Standard Model gauge kinetic terms. 8-10. `yt`, `yb`, `ytau`: third-generation Yukawas. 11-12. `lambdaH`, `mH2`: Higgs quartic and mass. 13-14. `lambdaPhi`, `mPhi2`: `Phi_c` quartic and mass. 15-16. `lambdaE`, `mE2`: `E` quartic and mass.
5. `kappaPhiE`: `Phi_c^2 E^2`. 18-19. `kappaHPhi`, `kappaHE`: `|H|^2 Phi_c^2`, `|H|^2 E^2`. 20-22. `xiH`, `xiPhi`, `xiE`: `R|H|^2`, `R Phi_c^2`, `R E^2`. 23-25. `Z_h`, `Z_phi`, `Z_E`: kinetic normalizations (projected as anomalous dimensions). 26-35. scalar dimension-six potentials: `|H|^6`, `Phi^6`, `E^6`, `|H|^4 Phi^2`, `|H|^4 E^2`, `|H|^2 Phi^4`, `|H|^2 E^4`, `Phi^4 E^2`, `Phi^2 E^4`, `|H|^2 Phi^2 E^2`. 36-38. derivative dimension-six operators: `|H|^2 Box |H|^2`, `Phi^2 Box Phi^2`, `E^2 Box E^2`. 39-41. gauge-strength portals: `|H|^2 F^2`, `Phi^2 F^2`, `E^2 F^2`.

Implementation names for entries 26-41 are, respectively: `c_H6`, `c_Phi6`, `c_E6`, `c_H4Phi2`, `c_H4E2`, `c_H2Phi4`, `c_H2E4`, `c_Phi4E2`, `c_Phi2E4`, `c_H2Phi2E2`, `c_HBox`, `c_PhiBox`, `c_EBox`, `c_HF2`, `c_PhiF2`, `c_EF2`.

Convention labels and fixed points (recorded, not guessed)

- Litim type-A Einstein-Hilbert benchmark (`gravity_frg.py`, hep-th/0606044, alpha -> infinity, g_eq = c_d gN with c_4 = 24 pi): fixed point `(lambdaN, gN) = (0.25, 1/(1536 pi))`, `eta_N* = -2`, `theta = 5/3 +- i sqrt(167)/3`.
- DEP gravity-matter block (`gravity_matter_coupling.py`, arXiv:1311.2898, type-II cut-off, alpha = rho = 1, table ghost factor 2*2^{[d/2]}): in the scaffold coordinates `gN = G_tilde` and `lambdaN = 2 Lambda_tilde`, so `beta_gN = beta_G_tilde` and `beta_lambdaN = 2 beta_Lambda_tilde`. Verified pure-gravity fixed point `(Lambda_tilde, G_tilde) = (0.01046, 0.7725)` and SM row `(-2.399, 1.762)`; the coupled SM joint fixed point in scaffold coordinates is `(gN, lambdaN) = (1.7621, -4.7975)` with the matter sector at its Gaussian point. These numbers are regulator/gauge dependent.

- The earlier claimed `(gN* ~ 0.27, lambdaH* ~ 0.038)` is **superseded**: it matches no recorded regulator, gauge, or projection convention, and it is not reproduced by any verified block above.
- Narain-Percacci nonminimal-coupling block (`scalar_tensor_nonminimal_np.py`, arXiv:0911.0386, convention label NP-DE-DONDER-TYPEA, d = 4, de-Donder gauge, optimized/Litim-shape cutoff, single scalar): the `xi_2` flow of the closed-form five-coupling system (Appendix A, eqs. A1-A5) is evaluated on the paper's GMFP gravity background `(lambda_0*, xi_0*) = (0.00862, 0.02375)` with `eta = 2`, using the per-species mapping `lambda_2 = m_i^2`, `xi_2 = xi_i`, `lambda_4 = lambda_i/4`. Verified against the paper: GMFP fixed point, critical exponents `2.143 +- 2.879 i`, and eigenvector `(0.3557 +- 0.3776 i, 0.8549)` (numerical Jacobian). Recorded paper-internal inconsistency: the printed closed-form entries of the stability matrix `M_00` are not the numerical derivatives of the Appendix A flows at the GMFP (the flow-derived Jacobian reproduces the published exponents and eigenvectors; the `M_00` closed forms do not). Note this block is a different gauge/cutoff convention from the DEP block; the two are never mixed in one solve (the `xi` entries sit on the NP GMFP background by construction).
- Sixth-order scalar LPA block (`sextic_lpa.py`): the same Litim-regulated Wetterich trace as the mass-squared block, projected at zero field onto the ten scalar dimension-six operators of the basis. Convention: every monomial `(h^2)^a (Phi^2)^b (E^2)^c` with `a+b+c = 3` carries a factor `1/2^(a+b+c)` (the same "2 per invariant factor" rule that fixes the quartic sector `lambda_H/4 (h^2)^2`, `p/4 (h^2) Phi^2`); radial projections then read `c_H6 = d^6 flow/dh^6 / 90`, `c_H4Phi2 = d^4 d^2 flow/(dh^4 dPhi^2) / 6`, `c_H2Phi2E2 = d^2 d^2 d^2 flow/(dh^2 dPhi^2 de^2)`, with the O(4) combinatorics fixed by the sextic tensor `c_000000 = 90 c_H6`, `c_0000PhiPhi = 6 c_H4Phi2`, `c_00PhiPhiEE = c_H2Phi2E2`. Scalar loops only; step choice `step = 5e-2` gives a convergence plateau (values at step 3e-2 and 7e-2 agree to a few percent).
- With the DEP SM gravity fixed point and Gaussian matter (all masses, quartics, Yukawas, `xi`, and dimension-six couplings zero), every currently derived entry vanishes: the `xi_i` entries exactly (verified residual ~1e-16) and the dimension-six entries to the stencil noise floor (~1e-11).
- Joint fixed point and 33-dimensional stability spectrum (`joint_fixed_point_stability.py`): the joint fixed point is the DEP SM fixed point `(lambdaN, gN) = (-4.7975095, 1.7621271)` (table convention, `improved=False`, ghost factor `2*2^{[d/2]}`) with all 31 other coordinates zero. The numerical Jacobian of the derived flow (quantum + canonical terms for `m_i^2` and the `c_*` entries) gives 33 critical exponents `theta = -eigenvalue`: **8 relevant** (`+3.9609`, `+2.8554 x3` for `mH2/mPhi2/mE2`, `+2.0000 x3` for `xiH/xiPhi/xiE`, `+1.6436`), **12 marginal** (`0.0000 x12`: the gauge, Yukawa, and six quartic couplings of the SM block, all Gaussian), **13 irrelevant** (`-0.0756 x3` from `eta_s = -0.076`, `-2.0000 x10` the dimension-six entries). The two gravity eigenvalues reproduce the DEP table exponents (3.961, 1.644).
- **R^2-sector record (negative result, not papered over)**: in the Codello-Percacci-Rahmede one-loop block `beta_{a_C2} = 133/(320 pi^2) ~ +0.04211` is strictly positive and constant, so the Weyl-squared direction runs away and this block has no fixed point for `a_C2`. `beta_{a_R2}` vanishes on the two lines `a_R2 = 2 x a_C2` with `x in {0.0295207, 0.4704793}`, but because `a_C2` never stops the `(a_R2, a_C2)` subsystem has no true fixed point either. `a_R2/a_C2` are therefore NaN at the Gaussian point (singular parametrization) and are excluded from the stability spectrum.
- The remaining undriven coordinates `c_HBox`, `c_PhiBox`, `c_EBox` (derivative dimension-six operators, not projectable by a constant-background LPA), and `c_HF2`, `c_PhiF2`, `c_EF2`

(gauge-strength portals, require a dressed-gauge-sector trace) are recorded as **pending**; no flow value is fabricated for them.

Exclusions and controls

- Four-derivative scalar operators are dimension eight and are excluded from this dimension-six basis, consistent with the primary-source correction.
- Redundant operators must be removed only by the declared integration-by-parts, equation-of-motion, and field-redefinition convention.
- Every result must include the operator table, gauge parameters, regulator, projection background, numerical residual `max_i |beta_i|`, and the full stability spectrum.

First non-negotiable checks

1. `beta_gN=(2+eta_N)gN`; nonzero `gN*` requires `eta_N*=-2`. This holds for the Litim type-A convention; under the DEP convention label the entries 0/1 are instead `beta_gN = beta_G_tilde` and `beta_lambdaN = 2 beta_Lambda_tilde` with `eta_h ~ 0.27 != -2` at the fixed point.
2. A proposed point must make every one of the 41 beta residuals small.
3. Critical exponents are the eigenvalues of the declared 41x41 Jacobian, with the sign convention stated alongside the result.